```
!pip install nltk spacy
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
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Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
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Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
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Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 56.3 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
import nltk
import spacy
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer
from nltk import pos_tag, ne_chunk

nltk.download('punkt')
nltk.download('stopwords')
nltk.download('averaged_perceptron_tagger')
nltk.download('maxent_ne_chunker')
nltk.download('words')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger.zip.
[nltk_data] Downloading package maxent_ne_chunker to
[nltk_data]     /root/nltk_data...
```

```
[nltk_data]   Unzipping chunkers/maxent_ne_chunker.zip.
[nltk_data] Downloading package words to /root/nltk_data...
[nltk_data]   Unzipping corpora/words.zip.
True
```

```
text = """I am learning Natural Language Processing.
It helps in understanding human language.
I like working on machine learning projects.
Text preprocessing is very important in NLP.
Tokenization splits sentences into words.
Stopword removal removes common words.
Stemming reduces words to base form.
POS tagging identifies grammar of words.=
Chunking forms meaningful phrases.
Dependency parsing shows word relations."""
```

```
import nltk
nltk.download('punkt_tab')
tokens = word_tokenize(text)
print("Tokenization:\n", tokens)
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
Tokenization:
 ['I', 'am', 'learning', 'Natural', 'Language', 'Processing', '.', 'It', 'helps', 'in', 'understanding', 'human', 'language', '.
```

```
stop_words = set(stopwords.words('english'))
filtered_tokens = [w for w in tokens if w.lower() not in stop_words]
print("\nAfter Stopword Removal:\n", filtered_tokens)
```

```
After Stopword Removal:
 ['learning', 'Natural', 'Language', 'Processing', '.', 'helps', 'understanding', 'human', 'language', '.', 'like', 'working', '
```

```
ps = PorterStemmer()
stemmed_words = [ps.stem(w) for w in filtered_tokens]
print("\nStemmed Words:\n", stemmed_words)
```

```
Stemmed Words:
 ['learn', 'natur', 'languag', 'process', '.', 'help', 'understand', 'human', 'languag', '.', 'like', 'work', 'machin', 'learn',
```

```
import nltk
nltk.download('averaged_perceptron_tagger_eng')
pos_tags = pos_tag(filtered_tokens)
print("\nPOS Tags:\n", pos_tags)
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.

POS Tags:
 [('learning', 'VBG'), ('Natural', 'NNP'), ('Language', 'NNP'), ('Processing', 'NNP'), ('.', '.'), ('helps', 'VBZ'), ('understan
```

```
grammar = "NP: {<DT>?<JJ>*<NN>}"
chunk_parser = nltk.RegexpParser(grammar)
chunk_tree = chunk_parser.parse(pos_tags)
print("\nChunking (Noun Phrases):")
print(chunk_tree)
```

```
Chunking (Noun Phrases):
(S
  learning/VBG
  Natural/NNP
  Language/NNP
  Processing/NNP
  ./.
  helps/VBZ
  understanding/VBG
  (NP human/JJ language/NN)
  ./.
  like/IN
  working/VBG
```

```
  (NP machine/NN)
  learning/VBG
  projects/NNS
  ./.
  Text/NNP
  preprocessing/VBG
  important/JJ
  NLP/NNP
  ./.
  Tokenization/NNP
  splits/VBZ
  sentences/NNS
  words/NNS
  ./.
  Stopword/NNP
  (NP removal/NN)
  removes/VBZ
  common/JJ
  words/NNS
  ./.
  Stemming/VBG
  reduces/NNS
  words/NNS
  (NP base/NN)
  (NP form/NN)
  ./.
  POS/NNP
  tagging/VBG
  identifies/NNS
  (NP grammar/NN)
  words/NNS
  ./.
  Chunking/VBG
  forms/NNS
  meaningful/JJ
  phrases/NNS
  ./.
  Dependency/NNP
  parsing/VBG
  shows/NNS
  (NP word/NN)
  relations/NNS
  ./.)
```

```
nlp = spacy.load("en_core_web_sm")
doc = nlp(text)

print("\nDependency Parsing:")
for token in doc:
    print(token.text, "->", token.dep_, "->", token.head.text)
```

```
 -> dep -> .
Text -> compound -> preprocessing
preprocessing -> nsubj -> is
```

```
form -> dobj -> base
. -> punct -> reduces

 -> dep -> .
POS -> compound -> tagging
tagging -> compound -> identifies
identifies -> compound -> grammar
grammar -> ROOT -> grammar
of -> prep -> grammar
words -> pobj -> of
. -> punct -> grammar

 -> dep -> .
Chunking -> ROOT -> Chunking
forms -> dobj -> Chunking
meaningful -> amod -> phrases
phrases -> dobj -> Chunking
. -> punct -> Chunking

 -> dep -> .
Dependency -> compound -> parsing
parsing -> nsubj -> shows
shows -> ROOT -> shows
word -> compound -> relations
relations -> dobj -> shows
. -> punct -> shows
```

Start coding or generate with AI.